

Customer Churn Prediction & LTV Engine

Week 4 • Day 5

Technical Documentation

Write professional README, architecture diagrams, and code documentation

■ Day Objectives

- ◆ Write a comprehensive README.md for the GitHub repository
- ◆ Create a SYSTEM_ARCHITECTURE.md with data flow diagrams
- ◆ Add docstrings to all Python modules
- ◆ Create a CHANGELOG.md documenting all 28 days of work
- ◆ Ensure all code follows PEP 8 style guidelines

■ 1. Professional README Structure

A great README is the front door of your project. It should answer: What is this? Why does it matter? How do I run it? What are the results?

■ 2. README.md Template

■ Complete README.md

```
# Customer Churn Prediction & LTV Engine
[![Python](https://img.shields.io/badge/Python-3.11-blue)]()
[![FastAPI](https://img.shields.io/badge/FastAPI-0.108-green)]()
```

```
[![PostgreSQL](https://img.shields.io/badge/PostgreSQL-15-blue)]()
```

Overview

A production-level ML system that predicts customer churn probability and calculates Customer Lifetime Value for telecom businesses. Built in 28 days as part of Zaalima Development Data Analytics Program.

Key Results

Metric	Value
XGBoost AUC-ROC	~0.86
Month-to-month churn rate	42.7%
High-value customers at risk	~TBD
API response time	<200ms

Architecture

Raw CSV → PostgreSQL → Pandas EDA → Feature Engineering → XGBoost Model → FastAPI → Metabase Dashboard

Quick Start

```
```bash
git clone https://github.com/YOUR_USERNAME/customer-churn-ltv-engine
cd customer-churn-ltv-engine
pip install -r requirements.txt
cp .env.example .env # edit with your DB credentials
python src/db/load_data.py
uvicorn src.api.main:app --reload
```
```

Tech Stack

- Python 3.11 | PostgreSQL 15 | FastAPI | Docker
- Pandas | Scikit-Learn | XGBoost | SHAP
- Metabase (Dashboard)

Project Structure

See SYSTEM_ARCHITECTURE.md for detailed architecture.

Author

Zaalima Development Data Analytics Program – 28-Day Project

■ 3. Commit & Checklist

- README.md updated with results table, quick start, architecture
- All Python files have module-level docstrings
- CHANGELOG.md documents all 28 days

■ Week4-Day5 Commit

```
git add .
git commit -m "Week4-Day5: Technical documentation – README, architecture, docstrings, CHANGELOG"
git push origin main
```

■ Estimated Time: 5 – 6 hours

— End of Week 4 Day 5 — Zaalima Development Confidential —